ANSWERS

1. MATRICES AND DETERMINANTS

Exercise 1.1

1.(i) $M_{11} = -1 \quad M_{12} = 0 \quad M_{21} = 20 \quad M_{22} = 5$

$A_{11} = -1 \quad A_{12} = 0 \quad A_{21} = -20 \quad A_{22} = 5$

(ii) $M_{11} = -12 \quad M_{12} = 2 \quad M_{13} = 23 \quad M_{21} = -16 \quad M_{22} = -4 \quad M_{23} = 14$

$M_{31} = -4 \quad M_{32} = -6 \quad M_{33} = 11$

$A_{11} = -12 \quad A_{12} = -2 \quad A_{13} = 23 \quad A_{21} = 16 \quad A_{22} = -4 \quad A_{23} = -14$

$A_{31} = -4 \quad A_{32} = 6 \quad A_{33} = 11$

2. 20 3. $\frac{1}{2}$ 4. –30 5. $\frac{13}{2}, 2$ 6. 0

Exercise 1.2

1. $\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$ 2. $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ 3.(i) $\frac{1}{5}\begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}$ (ii) $\frac{1}{10}\begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$

(iii) $\frac{1}{10}\begin{bmatrix} 10 & -10 & 2 \\ 0 & 5 & -4 \\ 0 & 0 & 2 \end{bmatrix}$ (iv) $\frac{1}{151}\begin{bmatrix} -22 & -46 & -7 \\ -13 & 14 & -11 \\ 5 & -17 & -19 \end{bmatrix}$

7. $\frac{1}{9}\begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -7 \\ 3 & -1 & -1 \end{bmatrix}$ 10. $\lambda = \frac{7}{4}$ 11. $p = 2, q = -3$

Exercise 1.3

1. $x = 1, y = 1$ 2.(i) $x = 3, y = -2, z = 1$

(ii) $x = -1, y = 2, z = 3$ (iii) $x = 1, y = -1, z = 2$ 3. 17.95, 43.08, 103.85

4. 3000, 1000, 2000 5. 13,2,5 6. 700, 600, 300

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